Python for Geospatial – Part 2: Python for Geospatial

(30 MINUTES)

AGAIF 2026 & Bootcamp FD1003

This session introduces raster-based geospatial analysis using Python. Learners will explore the fundamental concepts of raster data, understand the differences between vector and raster data models, and learn how to use Rasterio for reading, processing, and analyzing raster datasets. The workshop also introduces Folium for creating interactive web maps and demonstrates how Python provides an efficient environment for geospatial data science workflows. Through practical examples and demonstrations, participants will gain hands-on experience in managing raster data and visualizing geospatial information using Python.

Learning objectives

After completing this module, you will be able to:

- Explain the characteristics of raster data and differentiate raster and vector data models in GIS applications.
- Use Rasterio to read, visualize, process, and analyze raster datasets in Python.
- Create interactive web maps using Folium and implement geospatial workflows within a Jupyter Notebook environment.

1. Introduction

Raster data is one of the two primary methods used to represent geographic information in Geographic Information Systems (GIS). Unlike vector data, which stores geographic features as individual points, lines, and polygons, raster data represents the Earth's surface as a continuous grid of cells known as pixels.

Each pixel contains a value that describes a particular attribute of the geographic area covered by that cell. The value could represent elevation, temperature, rainfall, vegetation density, land cover type, soil moisture, or spectral reflectance recorded by a satellite sensor.

For example, consider a Digital Elevation Model (DEM). Each pixel stores the elevation value of that location above sea level. When viewed collectively, thousands or millions of pixels form a continuous representation of the terrain. Raster data is especially useful for representing phenomena that vary continuously across space, such as:

- Surface temperature
- Air pollution concentration
- Rainfall distribution
- Vegetation health
- Population density
- Satellite imagery

Because raster data captures continuous surfaces, it is widely used in remote sensing, environmental monitoring, agriculture, disaster management, and urban planning.

2. Components of Raster Data?

- A raster dataset consists of several key components.

Cell (Pixel)
The smallest unit of a raster dataset. Each pixel stores a value representing a characteristic of the Earth's surface.

Example:

Pixel	Elevation (m)
A1	120
A2	125
A3	130

Resolution
Resolution refers to the ground area represented by each pixel.

For example:

Resolution	Ground Coverage
1 m	Each pixel represents 1m × 1m
10 m	Each pixel represents 10m × 10m
30 m	Each pixel represents 30m × 30m

Higher resolution datasets contain more detail but require larger storage and processing power.

For instance:

A 1-meter resolution drone image may reveal individual trees, while a 30-meter Landsat image only captures larger land-cover patterns.

Bands
Raster datasets may contain multiple layers called bands.

Satellite imagery typically consists of several spectral bands.

Example:

Band	Spectral Region
Band 1	Blue
Band 2	Green
Band 3	Red
Band 4	Near Infrared

Different bands capture different characteristics of the Earth's surface.

The Near Infrared (NIR) band is particularly useful for vegetation analysis because healthy vegetation strongly reflects infrared radiation.

Real-World Applications of Raster Data

Agriculture

Raster imagery helps farmers monitor crop health using vegetation indices such as NDVI.

Benefits include:
- Detecting crop stress
- Optimizing fertilizer application
- Monitoring irrigation efficiency

Environmental Monitoring

Government agencies use raster datasets to monitor:

- Deforestation
- Coastal erosion
- Flood extent
- Forest fires

Satellite imagery allows large geographic areas to be monitored efficiently over time.

Urban Planning

Raster data supports:

- Land-use analysis
- Urban expansion monitoring
- Heat island assessment
- Infrastructure planning

2. Raster vs Vector Data

Understanding the differences between raster and vector data is fundamental for geospatial analysis because each data model is designed for different types of spatial information.

Vector Data

Vector data stores geographic features as discrete objects.

For example:
- A hospital is represented as a point.
- A road is represented as a line.
- A land parcel is represented as a polygon.
- Each feature can have associated attributes stored in a table.

Example:

School_ID	School Name	Students
101	ABC School	1200

Vector data is ideal when geographic boundaries must be represented precisely.

Raster Data

Raster data stores information as a continuous grid. Instead of storing the exact shape of a forest, raster data records the characteristic of each cell covering the forest area. This approach is particularly effective for representing naturally varying phenomena.

Examples:

- Temperature
- Elevation
- Rainfall
- Vegetation

Choosing Between Raster and Vector

A common misconception is that one model is better than the other. In reality, each serves a different purpose.

Use Vector When:
- Mapping roads
- Managing utility networks
- Storing property boundaries
- Performing network analysis

Use Raster When:
- Processing satellite imagery
- Terrain modelling
- Climate analysis
- Environmental monitoring

In practice, many GIS projects integrate both raster and vector datasets.

For example:

A flood risk study may use:
- Raster DEM data for terrain analysis
- Vector river networks
- Vector building footprints
- Raster rainfall data

Combining these datasets enables more comprehensive spatial analysis.

3. Introduction to Rasterio

Why Rasterio?

Before Python geospatial libraries became popular, many raster operations were performed using desktop GIS software such as:

- ArcGIS
- QGIS
- ERDAS Imagine
- ENVI

While these tools remain valuable, modern geospatial workflows increasingly rely on Python automation to improve efficiency and reproducibility. Rasterio was developed to simplify raster processing within Python. It provides a user-friendly interface built on top of the Geospatial Data Abstraction Library (GDAL), one of the most widely used geospatial software libraries in the world.

Advantages of Rasterio

Simplicity
Rasterio uses Pythonic syntax that is easier to understand than raw GDAL commands.

Integration
Works seamlessly with:

- NumPy
- Matplotlib
- GeoPandas
- Shapely

Automation
Enables large-scale processing of hundreds or thousands of raster files.

Reproducibility
Processing workflows can be documented and repeated consistently. This is particularly important in research and government projects where analyses must be transparent and repeatable.